



GS FOUNDATION 2024-25

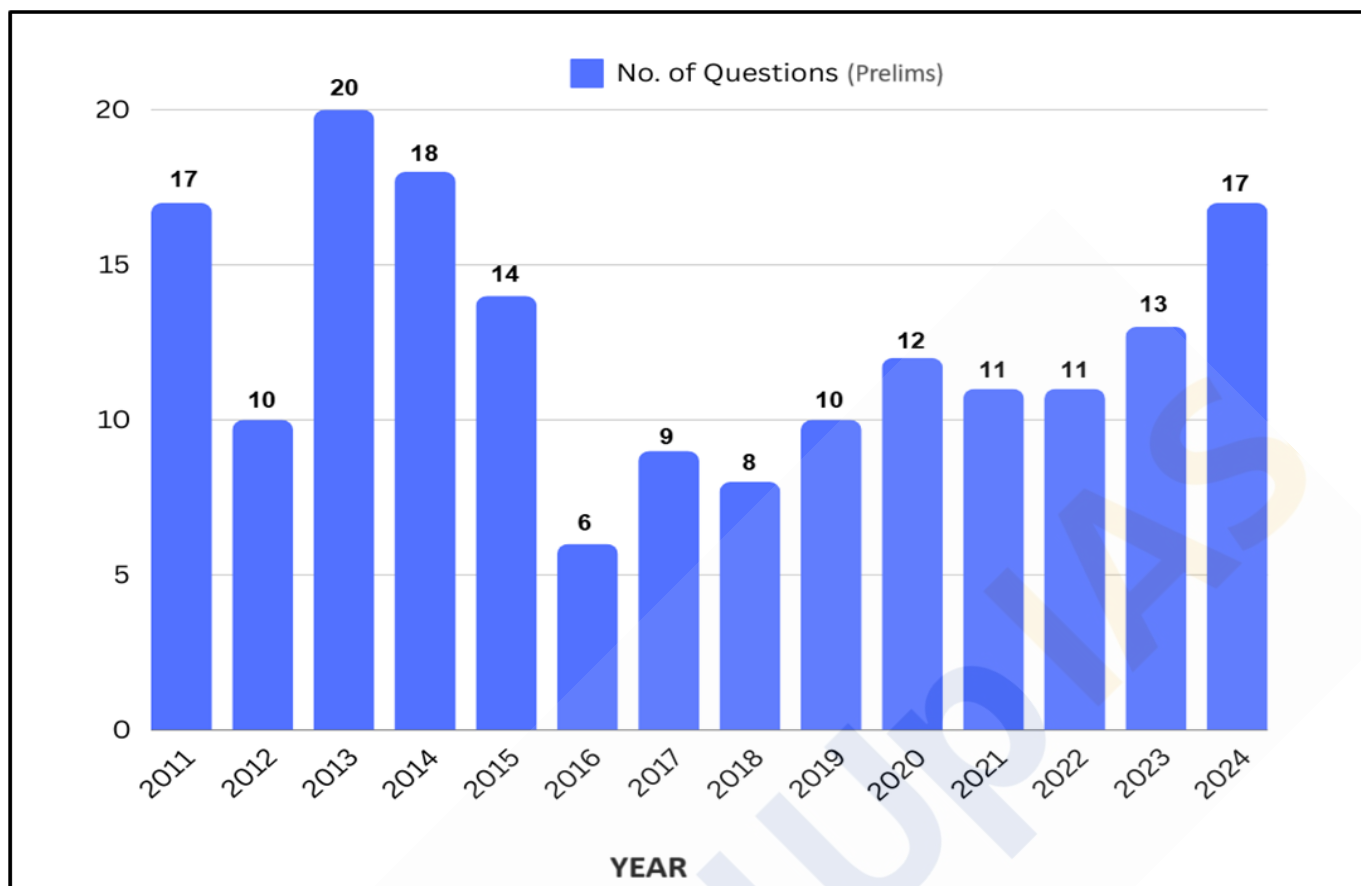
GEOGRAPHY - 7

Water in the Atmosphere - I

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Year-wise Trend Analysis (2011 – 2024)



Topic-wise Trend Analysis (2011 – 2024)

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Water in the Atmosphere

Introduction:

Water vapour is a **highly variable component** of the atmosphere. Its proportion varies from **zero to four percent** by volume of the atmosphere. Water is present in the atmosphere in **three forms namely gaseous, liquid, and solid**. The moisture in the atmosphere is derived from water bodies through evaporation and from plants through transpiration. There is a **continuous exchange of water** between the atmosphere, the oceans and the continents through the processes of evaporation, transpiration, condensation and precipitation.

Sources of Atmospheric Moisture: The Hydrologic Cycle:

The hydrologic cycle refers to the continuous movement of water within the Earth system, cycling endlessly between the atmosphere, soil, surface storage, lakes and streams, plants and animals, glacial ice, and the principal reservoir—the oceans. This cycle plays a crucial role in maintaining the water-vapour content in the atmosphere and is integral to weather phenomena and the distribution of moisture across the Earth's surface.

Components of the Hydrologic Cycle:

- ✓ **Evaporation:** Water evaporates from large and small water bodies, such as oceans, lakes, rivers, and moist soil, into the atmosphere.
- ✓ **Transpiration:** Vegetation loses moisture through a process called transpiration.
- ✓ **Evapotranspiration:** The combined effect of evaporation and transpiration from vegetation-covered land areas is known as evapotranspiration.
- ✓ **Condensation:** Water vapour in the atmosphere condenses to form clouds.
- ✓ **Precipitation:** Moisture returns to the Earth's surface as precipitation (rain, snow, sleet, etc.).

Despite the average constant water-vapour content of the atmosphere, there are local variations due to seasonal and weather changes. The moisture lost through precipitation is replenished by continuous evaporation and transpiration, ensuring the atmosphere's water-vapour content remains balanced.

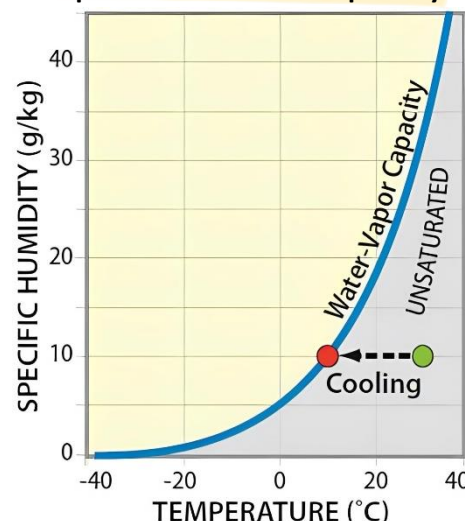
Humidity:

water vapour content at any time & place.

Humidity is a general term referring to the water vapour content of air at any one time and place. This term is not used to describe liquid droplets of fog, cloud, or rain. Since water vapour is dry like any other gas, it has nothing to do with dryness or wetness of the atmosphere. Although there is **spatial as well as temporal variation in the quantity** of water vapour present in the atmosphere, the **total amount of atmospheric water remains nearly constant**.

Humidity and Temperature:

- There exists a close balance between atmospheric humidity and air temperature. At any given point in time, the capacity of air to hold water vapor depends on its temperature. The moisture-holding capacity of air increases with rising temperatures. Notably, at higher temperatures, the capacity to hold moisture increases at a much faster rate. Conversely, at lower temperatures, this capacity increases slowly.



- The **moisture-holding capacity of air does not exceed a certain upper limit** despite large fluctuations in the amount of water vapor in the atmosphere. **This upper limit is determined by the temperature of the air.** The amount of water vapor that a given parcel of air can hold at a specific temperature is referred to as its capacity. This concept illustrates how air can hold more moisture as the temperature rises.
- At higher temperatures, the ability of air to hold moisture increases more rapidly.** This phenomenon is crucial in understanding weather patterns and climatic conditions, as it explains why humid conditions are often associated with warmer temperatures. Understanding the relationship between temperature and humidity is essential for predicting weather and studying climatic changes.

Humidity Measurements:

Absolute Humidity:

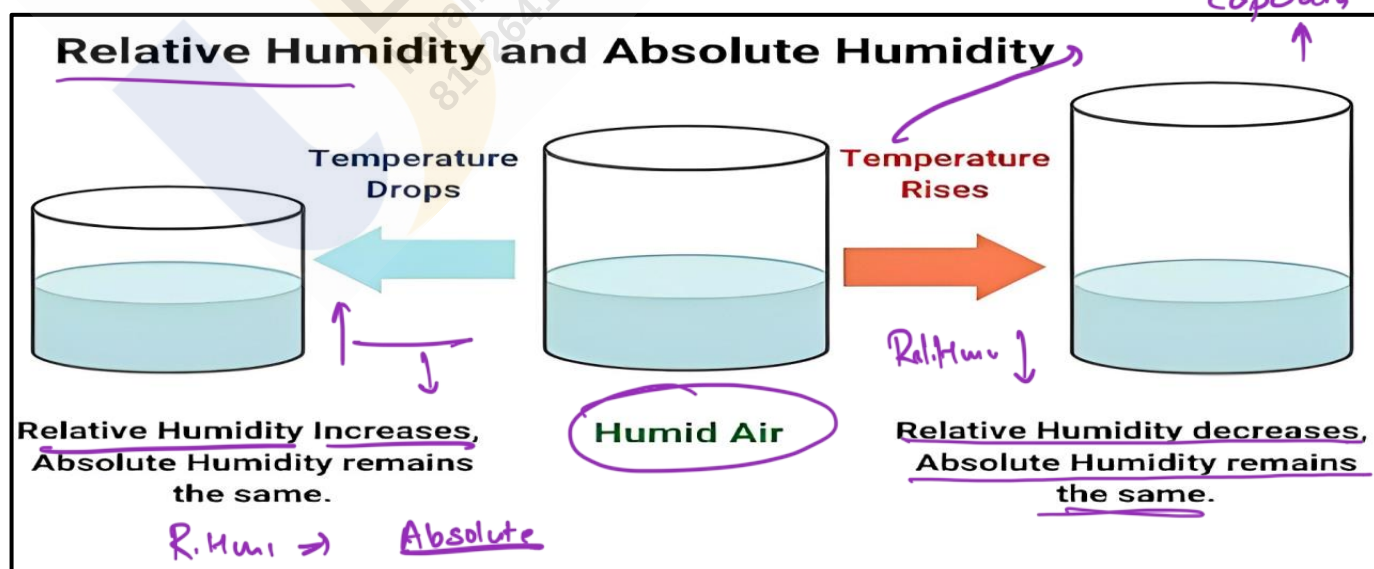
- The **actual amount of the water vapour present in the atmosphere** is known as the absolute humidity. It is the weight of water vapour per unit volume of air.
- It remains **relatively constant with temperature changes** unless additional water vapor is added or removed through evaporation or condensation.
- Absolute humidity **decreases from the equator towards the poles and from oceans to continents**, influencing the likelihood of precipitation. It is typically measured in grams per cubic meter of air and is closely related to vapor pressure.

Temp. is greater at equator than poles

*more ocean evaporation
→ so more continent.*

Relative Humidity:

- Relative humidity is the ratio of the amount of water vapor present in the air (absolute humidity) to the maximum amount of water vapor that the air can hold at a specific temperature and volume (humidity capacity), expressed as a percentage.
- It indicates **how close the air is to saturation with water vapor**. Relative humidity plays a critical role in weather forecasting and human comfort, with higher relative humidity levels indicating air that is close to being saturated with moisture.
- Relative Humidity = [Actual amount of water vapor in air (Absolute humidity)/Humidity at saturation point (the maximum water vapor air can hold at a given temperature)] X 100**



Importance of Relative Humidity (RH):

- **Climatic Significance** - Relative humidity indicates the likelihood of precipitation. **High RH suggests a greater possibility of wet (precipitation) conditions**, while **low RH indicates a tendency towards dry weather**. This makes RH a key factor in forecasting weather patterns and understanding climate variability.
- **Impact on Evaporation** - Evaporation rates are inversely related to relative humidity. **Higher RH reduces evaporation**, as the air is already saturated with moisture, whereas **lower RH enhances evaporation**, making it a critical factor in water management and agricultural practices.
- **Human Health and Comfort** - RH directly affects human health and comfort levels. **Very high (above 60%) and very low humidity levels are detrimental to human health, causing discomfort, respiratory issues, and heat-related illnesses**. Maintaining optimal RH levels is essential for indoor environments to ensure comfort and well-being.
- **Environmental Stability** - RH influences the stability of various materials, structures, and electronic devices. **High RH can lead to corrosion, mold growth, and damage to sensitive equipment**, while low humidity may cause static electricity and structural weaknesses in buildings.
- **Regional Implications** - Regions with high temperatures and **high RH, such as equatorial areas, pose health risks due to heat stress and humidity-related ailments**. Conversely, tropical hot deserts with **very low RH** levels create challenges related to **dehydration and environmental stressors**.

Evaporation:

Evaporation is the **process by which a liquid, such as water, transforms into gas or vapor**. It is a crucial part of the water cycle, where the atmosphere receives moisture from the Earth's surface through evaporation from oceans, lakes, rivers, and transpiration from plants.

During evaporation, **approximately 600 calories of energy are required to convert 1 gram of water into water vapor**. This energy is absorbed by water molecules from the surroundings, providing them with the kinetic energy needed to escape the liquid surface and become gas. **When water vapor condenses back into liquid form, this 'latent heat of vaporization' is released as heat energy**.

Evaporation is a **cooling process** because it **consumes energy from the surroundings**. As water evaporates, the remaining liquid is cooled by an equivalent amount of energy loss.

Factors Affecting Rate of Evaporation:

- **Amount of Water Available** - Evaporation rates are higher over oceans compared to continents due to the larger surface area of water available.
- **Temperature** - Higher temperatures increase the rate of evaporation as warmer air can hold more moisture, providing more energy for water molecules to transition to vapor.
- **Relative Humidity** - Evaporation rates decrease with higher relative humidity since moist air has a lower capacity to absorb additional moisture.
- **Wind Speed** - Wind enhances evaporation by dispersing moist air and maintaining a larger vapor-pressure difference between the evaporating surface and the atmosphere.
- **Area of Evaporating Surface** - Larger surface areas accelerate evaporation rates as more water is exposed to the air.

more evaporation
more sweating, more evaporation
more dehydration

- **Air Pressure** - Lower atmospheric pressure increases evaporation rates by reducing the pressure exerted on the evaporating surface.
- **Composition of Water** - Freshwater evaporates faster than saltwater due to differences in salinity affecting vaporization rates.

Condensation:

Condensation, the **process by which water vapor changes from a gas to a liquid**, is a fundamental part of the water cycle and plays a critical role in shaping weather patterns. This phenomenon occurs **when air containing water vapor cools down, causing the water molecules to slow down and come closer together**. Eventually, these attracted molecules form microscopic water droplets, visible as fog, dew, or clouds depending on the location and atmospheric conditions.

Understanding the Processes Leading to Condensation:

There are **two main categories** of processes that can trigger condensation: **Adiabatic and Non-adiabatic**.

Adiabatic Processes:

- These involve the **expansion and cooling of air without any heat exchange with the surroundings**. The most significant contributor to condensation in the free atmosphere is **this adiabatic cooling**.
- As **air rises, the pressure acting on it decreases**. This decrease in pressure causes the **air to expand**, and in accordance with the **principles of thermodynamics**, the expanding air performs work on its surroundings, using its own thermal energy. This **energy expenditure results in a decrease in the air's temperature, a phenomenon known as adiabatic cooling**.
- The rate of this cooling is typically around 10°C per km of ascent for unsaturated air. However, once the rising air reaches its condensation level, the latent heat released during condensation helps to counteract some of this cooling, reducing the rate to about 6°C per km.
- This modified rate is referred to as the **moist adiabatic lapse rate**. Convection currents, convergence of air masses along fronts, and orographic lifting (air forced upwards by mountains) are all examples of mechanisms that can cause adiabatic cooling and subsequent condensation.

Non-Adiabatic Processes:

- These involve the **transfer of heat between air and its surroundings**. While less impactful on widespread condensation in the atmosphere compared to adiabatic processes, non-adiabatic processes can **play a role in localized condensation events near the Earth's surface**. **Here's how they work:**
 - **Radiation:** When moist air loses heat directly to space through radiation, particularly during clear nights, the air temperature can decrease sufficiently to reach the dew point, initiating condensation. This radiative cooling is responsible for the formation of dew.
 - **Conduction:** If warm, moist air comes into direct contact with a cold surface, such as a body of water or the ground at night, heat transfer through conduction can occur. As the air loses heat to the colder surface, its temperature can drop below the dew point, leading to condensation in the form of dew or fog.
 - **Mixing:** When warm, moist air mixes with cooler air masses, the combined temperature may decrease to the dew point, triggering condensation. This can sometimes lead to the formation of fog.

It's important to note that **non-adiabatic cooling processes are generally confined to a thin layer of air near the Earth's surface** and are **limited in their ability to produce significant precipitation**. They are primarily responsible for the formation of dew, frost, and fog.

Necessary Conditions for Condensation:

Saturation:

- Air must be saturated with water vapor for condensation to occur. Saturation signifies that the air can no longer hold any additional water vapor in its gaseous state. This condition can be reached in **two ways**:
 - By cooling the air
 - By adding more moisture to it.

Condensation Nuclei:

- Even when air is saturated, **microscopic particles suspended in the air are needed for water vapor to condense upon**. These particles, known as **condensation nuclei or hygroscopic nuclei**, attract water molecules due to their chemical properties. Common examples of condensation nuclei include **sea salt particles, smoke particles, and dust particles**.
- **In the absence of these nuclei, much higher relative humidity levels (around 400%) would be required for condensation to initiate.** Fortunately, the atmosphere typically contains an abundance of hygroscopic nuclei, ensuring condensation can occur at much lower relative humidity levels (around 75%). As the relative humidity rises towards 100% saturation, condensed water droplets around these nuclei rapidly grow in size, eventually becoming visible as clouds.

Forms of Condensation:

Dew:

- When the **atmospheric moisture is condensed and deposited in the form of water droplets on cooler surface of solid objects such as grass blades, leaves of plants and trees and stones**, it is termed as dew. Condensation in dew form occurs when there is clear sky, little or no wind, high relative humidity and cold long nights. These conditions lead to greater terrestrial radiation and the solid objects become cold enough to bring the temperature of air down below dew point. In this process the extra moisture of the air gets deposited on these objects. Dew is formed when dew point is above freezing point. Dew formation can be seen if the water is poured into a glass from the bottle kept in a refrigerator. The outer cold surface of the glass brings the temperature of the air in contact with the surface down below dew point and extra moisture gets deposited on the outer wall of the glass.



Frost:

- When the **dew point is below freezing point, under above mentioned conditions, the condensation of extra moisture takes place in the form of very minute particles of ice crystals**. It is called frost. In this process, the air moisture condenses directly in the form of tiny crystal of ice. This form of condensation is disastrous for standing crops such as potato, peas, pulses, grams, etc. It also creates problems for road transport system.



Mist and Fog:

- When **condensation** takes place in the air near the **earth's surface in the form of tiny droplets of water hanging and floating in the air**, it is called mist. In mist the visibility is more than one Kilometer and less than two kilometers. But **when the visibility is reduced to less than one Kilometer**, it is called fog.
- **Ideal conditions for the formation of mist and fog are clear sky, calm and cold winter nights.**

**Haze:**

- Haze is traditionally an **atmospheric phenomenon where dust, smoke and other dry particles obscure the clarity of the sky** (No condensation. Smog is similar to haze but there is condensation in smog). Sources for haze particles include farming (ploughing in dry weather), traffic, industry, and wildfires.

Smog:

- Smog is a **fog that has been polluted and discoloured by smoke, dust, carbon monoxide, sulphur dioxide and other fumes**. Smog frequently occurs in large cities and industrial centres. It causes respiratory illness.

**Cloud:**

- Clouds are **visible aggregates of water droplets, ice particles, or a mixture of both along with varying amounts of dust particles**. Clouds are generally **classified on the basis of their general form or appearance and altitude**.

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